**Write a Python program to implement an Autoencoder for the Diabetes dataset. Load the Diabetes dataset, explore and preprocess the numerical features, handle any missing values if present, and standardize the input features. Construct an encoder–decoder neural network that learns a lower-dimensional latent representation of the diabetes-related features and reconstructs the original input data. Train the Autoencoder using the input features as both input and reconstruction target. Evaluate the reconstruction performance using MSE, MAE, and RMSE, visualize the original and reconstructed feature values, and experiment with different latent-space dimensions (e.g., 2, 4, and 8) to compare their reconstruction performance.**

https://www.kaggle.com/search?q=diabetes+dataset

https://www.kaggle.com/code/hassaan2580/california-housing-dataset

**Task 1: Data Loading**

- Number of rows and columns
- Column names
- Data types
- First five records
- Missing values

**Task 2: Data Preprocessing**

- Handle missing values, if present.
- Remove unnecessary columns.
- Select numerical features.
- Normalize the features using StandardScaler or MinMaxScaler.
- Divide the data into training and testing sets.

**Task 3: Design the Autoencoder**

Construct an autoencoder consisting of:

- Input layer
- At least two encoder layers
- A bottleneck/latent layer
- At least two decoder layers
- Output layer having the same number of neurons as the input features

**Task 4: Model Training**

Train the autoencoder using:

- Optimizer: Adam
- Loss function: Mean Squared Error (MSE)
- Suitable batch size
- At least 50 epochs

Use a validation set during training.
The model should learn to reconstruct the original input:

**Task 5: Performance Evaluation**

- Mean Squared Error (MSE)

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

Calculate the reconstruction error for individual samples as well.

**Task 6: Visualization**

Generate the following plots:

1. Training loss vs. validation loss
2. Original data vs. reconstructed data (scatter plot)
3. Distribution of reconstruction errors (histogram)